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EEE 202-1

LAB-05: Coupled Inductors and Mutual Inductance

Introduction:

The goal of this assignment is to understand the behavior of coupled inductors with normal and reversed coupled attributes. For the first part, the derivations required for verifying the results of LTSpice simulations will be made for both normal and reverse coupled inductors. After the required derivations have taken place, the circuits will be simulated using LTSpice with parameters that will be decided later on. The results of the simulation will be compared to our derivations to find an error between the two. By these steps, we will have constructed a strong basis to compare a real-life coupled inductor circuit to for further analysis during the hardware part. After these steps, windings will be wound around an inductor core and certain datapoints of the core will be gathered for different conditions. These results will be compared with an LTSpice simulation of the circuit to inspect the errors of the circuit.

Analysis:

Before taking on the preliminary section, the main equations that will be used during our derivations will be recalled. But before that, it should be noted that we were assigned a turn number of 25 in the Lab 5 Toroid Assignment document. It should be noted that we are given the equations:

$$V_1 = j\omega L_1 I_1 + j\omega M I_2, \qquad V_2 = j\omega L_2 I_2 + j\omega M I_1 \quad (1)$$

for two series coupled inductors with the same coupling direction. It should be noted that V_1 and V_2 denote voltages of the respective inductors, I_1 and I_2 denote the currents of the inductors, L_1 and L_2 denote the inductance of the respective inductors, ω is the input angular frequency of the source and M is the mutual inductance of the inductors. If the coupling direction is reversed, the set of equations become:

$$V_1 = j\omega L_1 I_1 - j\omega M I_2, \qquad V_2 = j\omega L_2 I_2 - j\omega M I_1 \quad (2)$$

with the same variables denoting the same things.

Additionally, as per the Lab 5 document, it should be noted that these circuits have equivalent forms that include ideal transformers. For the sake of completeness, these circuits are given (Figure A-B).

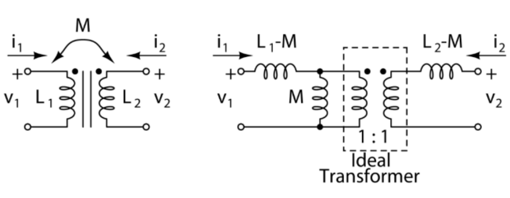

Figure A: Equivalent Circuit of Coupled Series Inductors

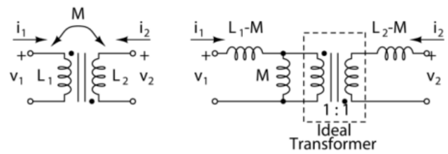

Figure B: Equivalent Circuit of Reverse Coupled Series Inductors

After these equivalencies were noted, the preliminary tasks were to be taken on.

Preliminary Part 1 (Series Coupled Inductance Derivation):

For a series set of coupled inductors, we know by Kirchoff's Current Law, the current through both inductors must be the same. Utilizing this fact and denoting the current by I and recognizing the net voltage will be of the net inductance, we can modify equation (1) to:

$$V_1 = j\omega L_1 I + j\omega M I, \qquad V_2 = j\omega L_2 I + j\omega M I \rightarrow V_{net} = V_1 + V_2 = j\omega I(L_1 + L_2 + 2M) \quad (3)$$

Since this analysis in the phasor domain, we can use Ohm's Law in order to get the impedance of the net inductance. Additionally, we can utilize the identity for the impedance of an inductor, which is $Z_L = j\omega L$, we can derive:

$$Z_{L_{net}} = \frac{V_{net}}{I} = j\omega(L_1 + L_2 + 2M) \rightarrow L_{net} = L_1 + L_2 + 2M \quad (4)$$

Which yields the net inductance of two coupled inductors in series.

Preliminary Part 2 (Series Reversed Coupled Inductance Derivation):

We can take the same steps we did in Preliminary Part 1 for the case for reverse coupled inductors. Again, by denoting net current as I and net voltage as V_{net}, we can rewrite equation (2) as:

$$V_1 = j\omega L_1 I - j\omega M I, \qquad V_2 = j\omega L_2 I - j\omega M I \rightarrow V_{net} = V_1 + V_2 = j\omega I(L_1 + L_2 - 2M) \quad (5)$$

And by taking advantage of the impedance formula for inductors, we get:

$$Z_{L_{net}} = \frac{V_{net}}{I} = j\omega(L_1 + L_2 - 2M) \rightarrow L_{net} = L_1 + L_2 - 2M \quad (6)$$

Which yields the net inductance of two reverse coupled inductor in series.

Preliminary Part 3 (Mutual Inductance Derivation):

It is said that the mutual inductance M can be found by the relation of the coupled series and reverse coupled series net inductances. If we denote the net inductance of the coupled series inductors by L_C and the net inductance of the reverse coupled series inductors by L_R, we can subtract the result of equation (6) from equation (4) to get:

$$L_C - L_R = 4M \rightarrow M = \frac{L_C - L_R}{4} \quad (7)$$

which is the mutual inductance in terms of the series net inductances for the coupled and reverse coupled case.

Preliminary Part 4 (Transformer Open Circuit Voltage):

In this part, we are asked to find the open circuit voltage of the secondary of a transformer (Figure C). This is due to the open circuit causing no current to flow and the generated voltage being equal across the inductor and the transformer's secondary. For this image, k is given by the relation $k = \frac{M}{\sqrt{L_1 L_2}}$.

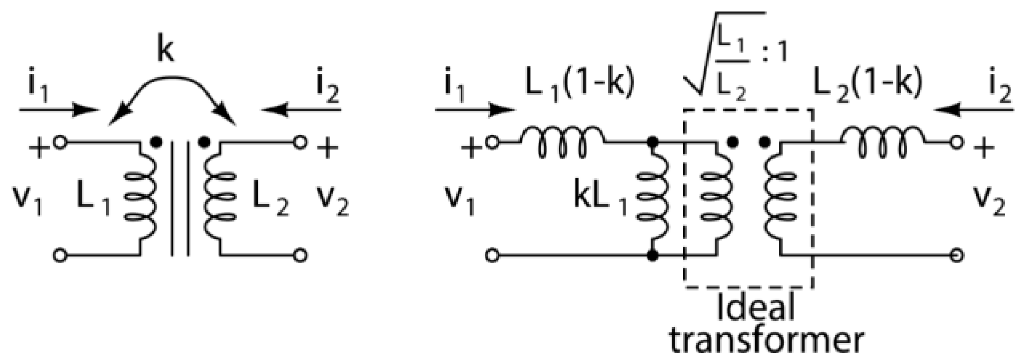

Figure C: Alternative Equivalent Circuit of Coupled Inductors

By treating the circuit at the primary as a voltage divider and not letting any current flow through the primary winding of the transformer since no current flows for the secondary when it is open circuited, we get the voltage divider formula:

$$V_{primary} = V_1 \frac{jkL_1}{jkL_1 + j(1-k)L_1} = V_1 k \quad (8)$$

Which is a simple relation dependent on k. Since we are asked to solve for the response for an unit voltage source, which is a source of voltage 1 for time greater than zero, we can say that the primary's voltage is equal to k for positive time. For the secondary voltage, we can utilize the property of transformers for voltage where $\frac{V_1}{V_2} = \frac{N_1}{N_2}$, such that N_1 and N_2 are the turn numbers for the primary and secondary respectively. By looking at Figure C, we see that this ratio is equal to $\sqrt{\frac{L_1}{L_2}}$, so we can write:

$$\frac{V_{primary}}{V_{secondary}} = \sqrt{\frac{L_1}{L_2}} \rightarrow V_{secondary} = k\sqrt{\frac{L_2}{L_1}} \quad (9)$$

Which is the voltage across the inductor L_2 in terms of L_1, L_2 and k.

Preliminary Part 5 (LTSpice Coupling Simulation):

For this part, we were to construct a circuit such that we were able to compare our values to our derivations in the first preliminary part. The values for L_1 and L_2 were both selected to be $66.25\ \mu H$, as this is the value that is equal to the turn number squared times the inductance per turn squared value of the core that will be used in the hardware section. To recall, we were assigned a turn number of 25 and the core has an inductance per turn squared value of $0.106\ \mu H/n^2$. The magnetic coupling coefficient was chosen to be 0.5. After these values were chosen, the design was created in LTSpice and the voltage across the inductors were plotted in accordance to the Lab 5 document (Figure D).

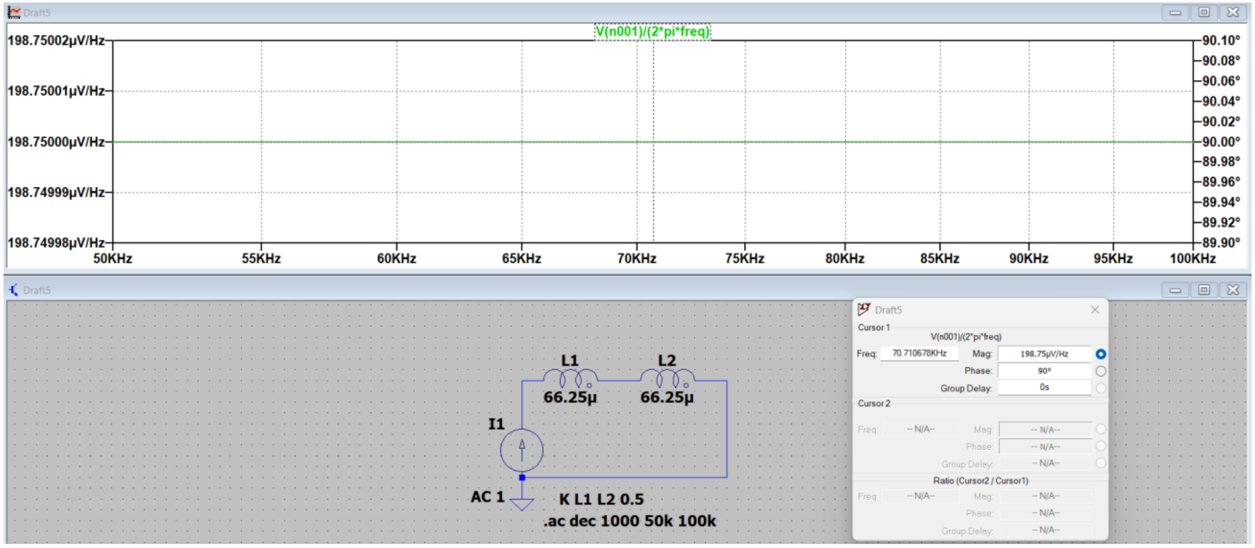

Figure D: LTSpice Circuit and Plot for the Coupled Inductors

By analyzing this plot by using a cursor, it was observed that the plot had a constant response of $198.75 \frac{\mu V}{Hz}$ for all frequencies in our assigned range. Since the used a source of amplitude 1A, by using Ohm's Law, it was found that the net inductance was $198.75\ \mu H$. Since these are inductors with standard coupling, we can utilize equation (4) to find the mutual inductance. Inserting our values, we get:

$$198.75\ \mu H = 66.25\ \mu H + 66.25\ \mu H + 2M \rightarrow M = 33.125\ \mu H \quad (10)$$

And by comparing this value for the equation of magnetic coupling, we have:

$$k = \frac{M}{\sqrt{L_1 L_2}} \rightarrow 0.5 = \frac{M}{\sqrt{(66.25\ \mu H)(66.25\ \mu H)}} \rightarrow M = 33.125\ \mu H \quad (11)$$

Since both of these values for the mutual inductance are exactly the same, we have verified that our result for part 1 of the preliminary section is correct.

Preliminary Part 6 (LTSpice Reverse Coupling Simulation):

For this part, the same procedures were taken as part 5 of the preliminary section but the second inductor was flipped to reverse the coupling of the inductors. The values of the components and frequency range were kept the same. The simulation was ran and the same point was probed according to the Lab 5 document (Figure E).

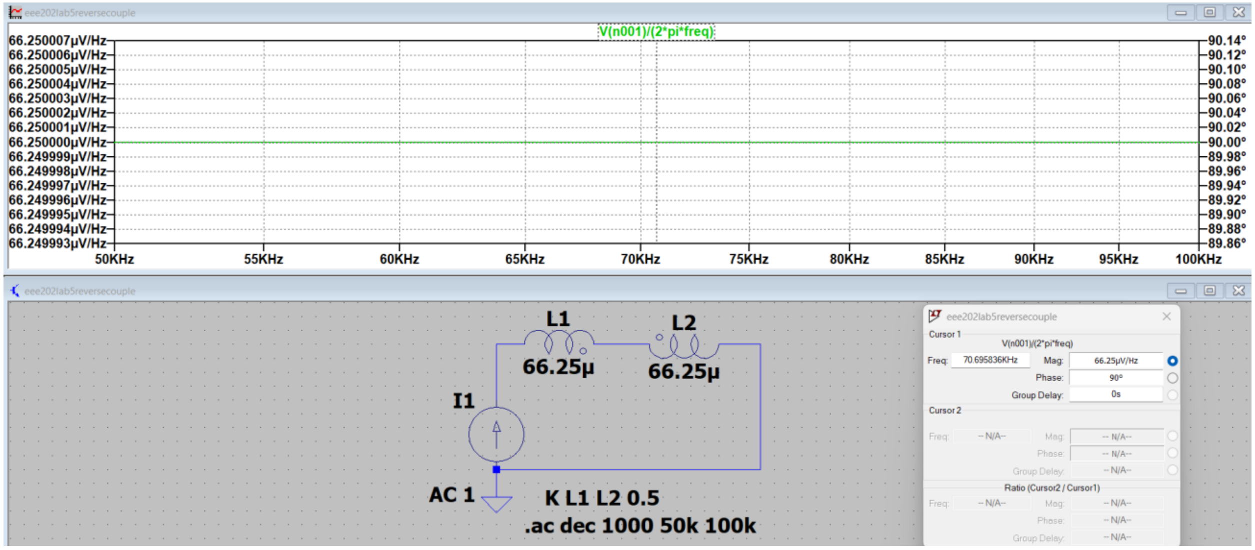

Figure E: LTSpice Circuit and Plot for the Reverse Coupled Inductors

By analyzing this plot by using a cursor, it was observed that the plot had a constant response of 66.25 $\frac{\mu V}{Hz}$ for all frequencies in our assigned range. Since the used a source of amplitude 1A, by using Ohm's Law, it was found that the net inductance was 66.25 μH. Since these are inductors with standard coupling, we can utilize equation (6) to find the mutual inductance. Inserting our values, we get:

$$66.25\ \mu H = 66.25\ \mu H + 66.25\ \mu H - 2M \rightarrow M = 33.125\ \mu H \quad (12)$$

This yields the same result we got in the fifth preliminary section, so our derivations for part 2 were verified to be true.

Preliminary Part 7 (LTSpice Reverse Coupling Simulation):

For this part, we were asked to verify the resultant equation in the third preliminary section. Recalling the values we gathered from the LTSpice simulations in part five and six and inserting them into equation (7), we can find the mutual inductance M to be:

$$M = \frac{L_C - L_R}{4} = \frac{198.75 - 66.25}{4}\ \mu H = 33.125\ \mu H \quad (13)$$

Since this value is equal to the value we got from the simulations in part five and six, the derivation in part three of the preliminary was also verified.

Preliminary Part 8 (Inherent Inductor Resistance):

For this part, firstly, we are asked to calculate the parasitic resistances for our inductors. This inherent resistance's value is given by the relation $r = \frac{\omega L}{Q}$, which was found in the Lab 5 document. Since we are asked to calculate the resistance for a test frequency of 100 kHz and a Q of 60, we can find that the resistance for our inductors to be approximately 0.6938 Ohms, due to the fact that both of our inductors have the same value. Through this value, an LTSpice circuit was configured in a way that was equivalent to the circuit in Figure C with additional series inductors. The circuit was simulated in accordance to the given parameters in the Lab 5 document and the secondary voltage was probed (Figure F).

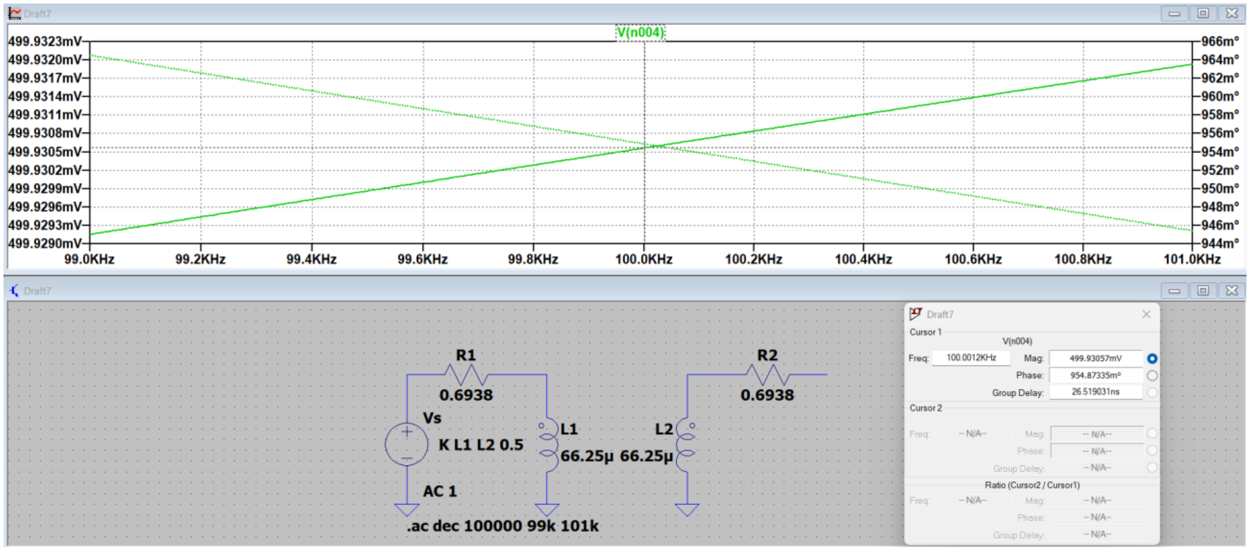

Figure F: LTSpice Circuit and Plot for the Coupled Inductors with Series Resistances

By analyzing the plot at our test frequency of 100 kHz, a voltage with the amplitude 0.4999 was observed. By inserting our component values into equation (9), we would normally expect the secondary voltage to be 0.5 Volts for a case with no resistors ($V_{secondary} = k\sqrt{\frac{L_2}{L_1}}$). This means that an error of 0.0200% was observed. This could be caused by the resistors loading the inductors to lessen the current received. However, since this error value is extremely small, we can say that small resistances are negligible for the response of coupled inductors. Additionally, the delay at our frequency was found to be 0.9549 degrees.

Preliminary Part 9 (Loaded Secondary Winding):

For the last preliminary part, a 10 Ohm resistor was added to the second inductor as per the instructions. After the resistor was added, the simulation was reran (Figure G).

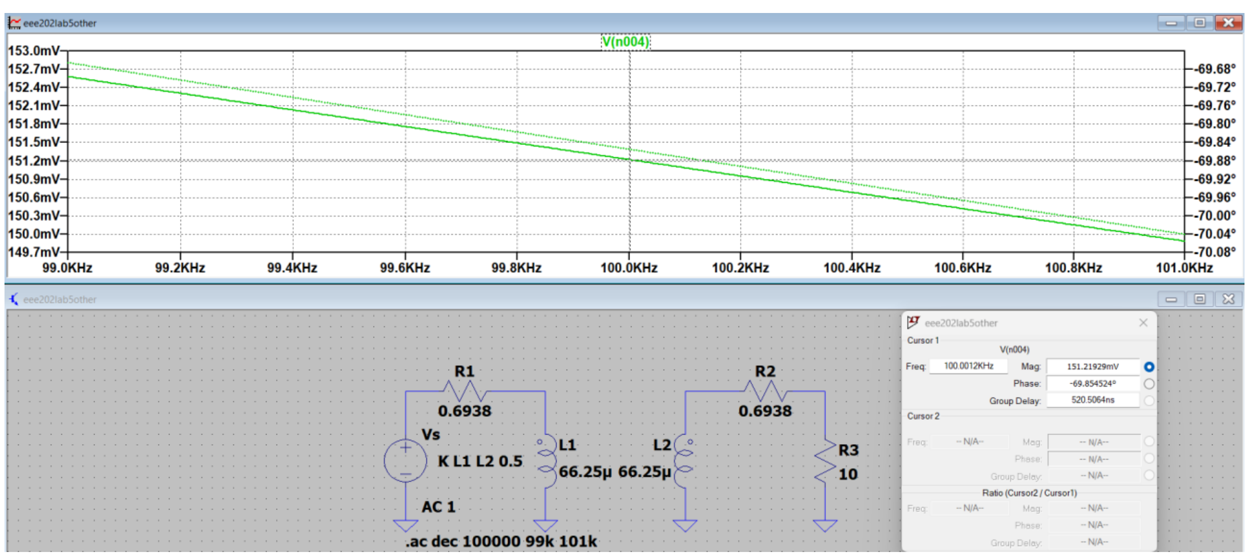

Figure G: LTSpice Circuit and Plot for the Coupled Inductors with Series Resistances and Load

When the plot was analyzed at the same frequency of 100 kHz, a voltage with an amplitude of 0.1512 and a phase of -69.8545 degrees was observed. A relatively high drop in voltage was observed. Compared to the previous part, for the voltage, an error of 69.7540% was calculated. This is due to the fact that, by loading the inductor L_2, we have now allowed current to flow, which will affect our measurements. The additional current changes the magnetic coupling factor k, and an additional drop happens due to the internal series resistance of the inductor interacting with the load resistor. The increase in phase can also be explained by these factors. The additional induced current will significantly alter the phase of the voltage, as there will always be a 90-degree delta between an inductors current and its voltage. Through all of these steps, the preliminary work section was done.

Hardware Implementation:

After the preliminary steps were taken, the circuit was to be built in real-life. Firstly, the assigned toroid core with the primary winding number of 25 was acquired. An additional copper wire was gathered and wound around the toroid core in order to achieve a winding number of 10. This number was chosen arbitrarily from the given range. The ends of both wires were sanded down to bare copper and tinned using unleaded solder for better connectivity. Now that the full coupled inductor was achieved, the core's inductance and quality factor values for L_1, L_2, coupled and reverse coupled combinations of L_1 and L_2 were measured using an RLC meter at a test frequency of 100 kHz (Figure 1.1-1.4).

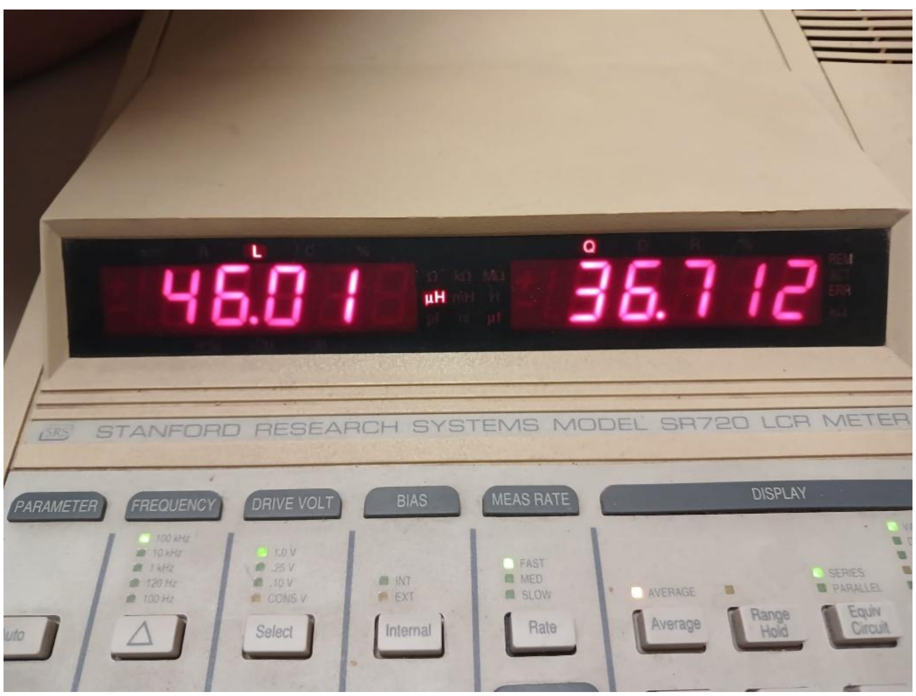

Figure 1.1: Inductance and Quality factor for L_1, $N_1 = 25$

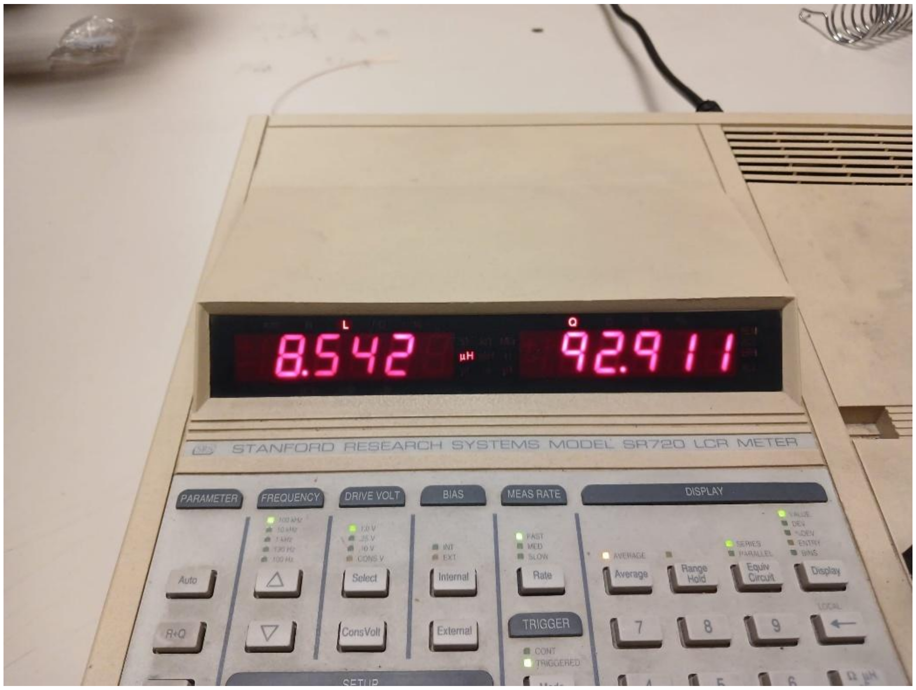

Figure 1.2: Inductance and Quality factor for L_2, $N_2 = 10$

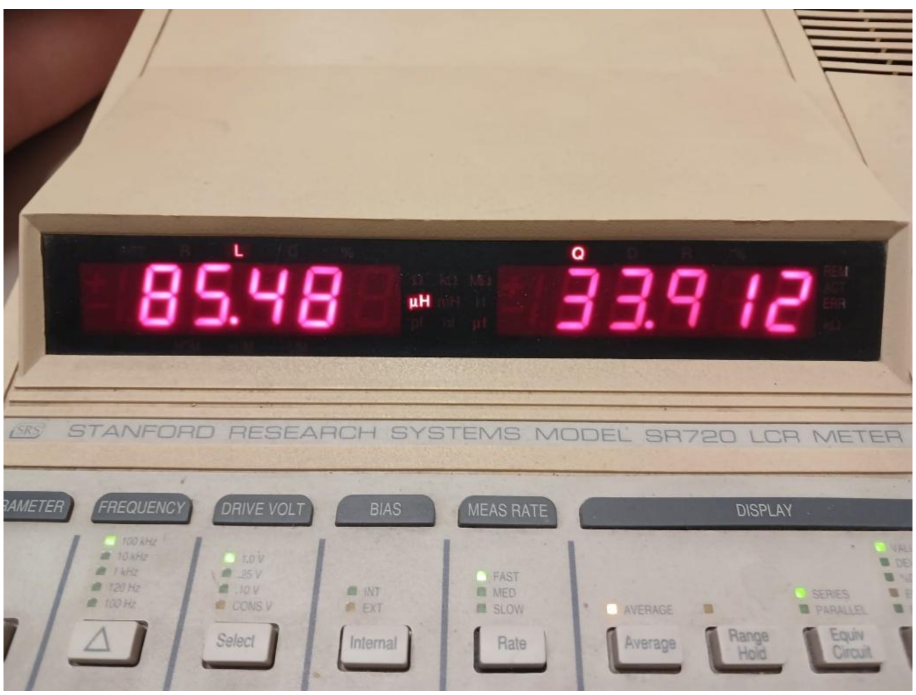

Figure 1.3: Inductance and Quality factor for the Coupled Series Inductors

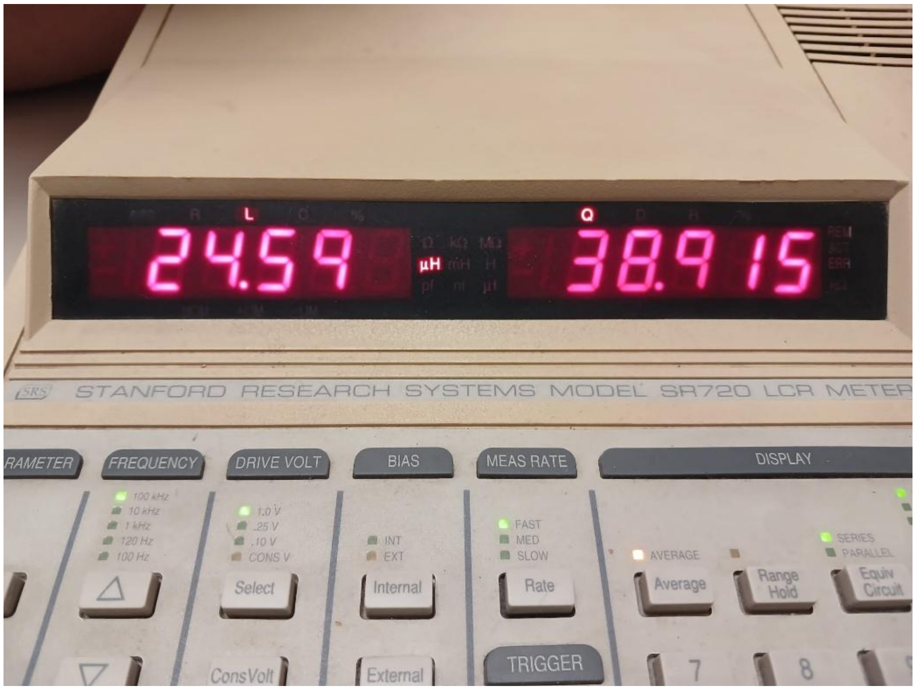

Figure 1.4: Inductance and Quality factor for the Reverse Coupled Series Inductors

By inspecting all of these values, it was seen that all of the quality factors were observed to be above 30, so the measurements were deemed acceptable. The inductance values for L_1 and L_2 were measured to be 46.01 µH and 8.542 µH respectively. With the given inductance per turn squared ratio of 106 nH/T², the ideal values for the turn numbers for 25 and 10 were calculated to be 62.25 µH and 10.6 µH respectively. By comparing these values, an error of 26.0884% and 19.4151% were found for L_1 and L_2 respectively. These error values can be caused by the windings of the inductor not being tight enough and being too far apart. However, since we were to update our LTSpice models in accordance to these measurements to compare the actual errors, these values were deemed acceptable. Using the quality factors for the respective lone inductors, using the formula $r = 2\pi f L/Q$ for the frequency of 100 kHz, their series resistances were calculated to be 0.7875 and 0.0578 for L_1 and L_2 respectively. Additionally, using the coupled and reverse coupled inductor values, 85.48 µH and 24.59 µH, in addition with equation (13), the mutual inductance M was found to be 15.2225 µH. Inserting the values for M, L_1 and L_2 into equation (11), the coupling factor k was found to be 0.768. All of these pieces of information were combined into the following tables in according to the Lab 5 document (Table 1-3).

	Measured (µH)	Measured Q	r (Ω)	$A_L N^2$ (µH)
L_1	46.01	36.712	0.7875	62.25
L_2	8.542	92.911	0.0578	10.6

Table 1: Information on individual L_1 and L_2

	Measured (µH)	Measured Q
Coupled Series Inductors	85.48	36.712
Reverse Series Inductors	24.59	38.915

Table 2: Information on Coupled and Reverse Coupled L_1 and L_2

	Measured (µH)
N_1	25
N_2	10
$M\ (\mu H)$	15.2225
k	0.768

Table 3: Miscellaneous Information of the Setup

After these datapoints were gathered, enough information to base an LTSpice simulation was gathered. Before that, a signal generator's output was connected to the legs of L_1 and the signal generator was set to 5 Volts peak-to-peak at a frequency of 100 kHz. The SYNC output of the signal generator was connected to the EXT TRIG port of the oscilloscope. The oscilloscope's parameters were set in accordance to the instructions in the Lab 5 document. After this, two oscilloscope probes were taken and connected to L_1 and L_2 of the toroidal coupled inductors with the signal generator on to measure V_1 and V_2 (Figure 2.1). The setup was such that, for this instance, CH1 displayed V_1 and CH2 displayed V_2 of our circuit. After making sure that the secondary winding of the inductor was kept an open circuit, the voltage and the delay between the voltages were observed (Figure 2.2-2.3).

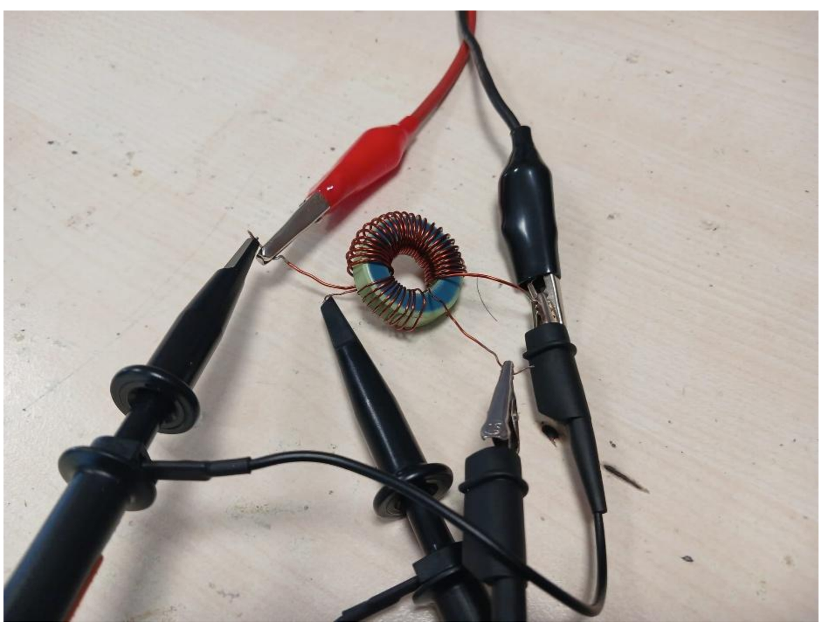

Figure 2.1: Implemented design with open circuited second winding

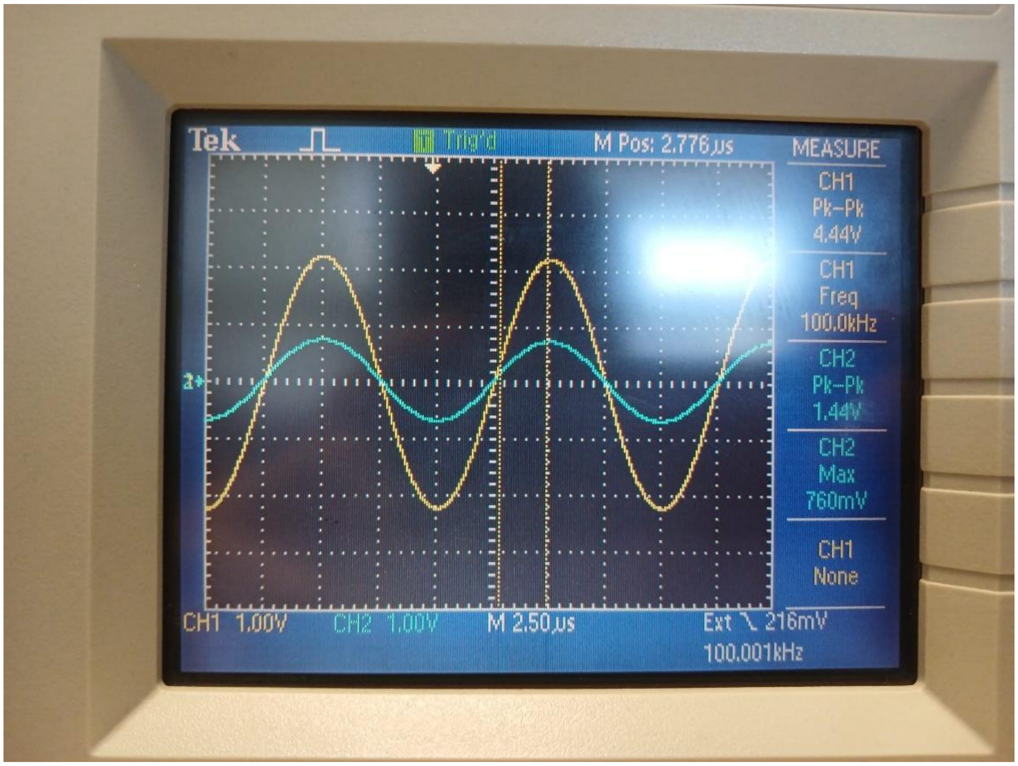

Figure 2.2: Voltages of L_1 $(CH1)$ and L_2 $(CH2)$

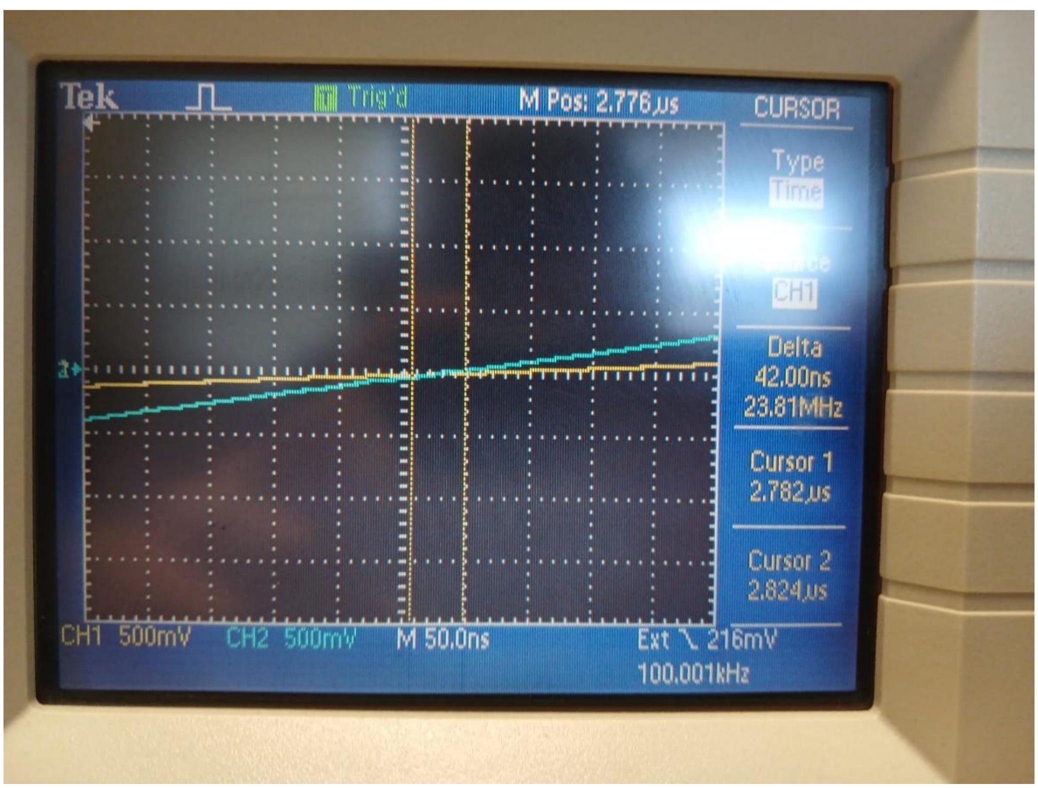

Figure 2.3: Delay between the probed voltages' waveforms

From these images, the peak-to-peak voltages were found to be 4.44 and 1.44 for V_1 and V_2 respectively. Additionally, the delay was found to be approximately 42 ns. When converting this value to degrees, it was multiplied by $\frac{360°}{10\mu s}$, which yielded 1.5120 degrees. These values were taken a note of before further analysis. By utilizing the data in Table 1, the LTSpice model in Figure F was updated. Thei input voltage was set to 5 Volts as an amplitude and a series source resistance of 50 Ohms were added. The simulation was ran and the voltages of the inductors were cursored at the points that included their series resistances (Figure 2.4).

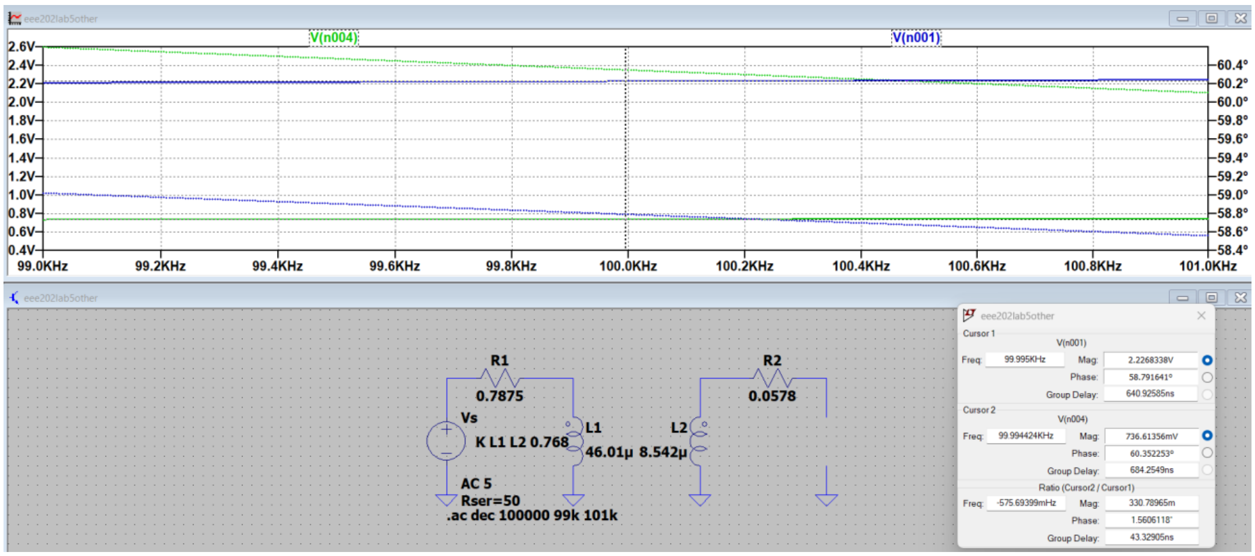

Figure 2.4: Simulation of the real-life circuit with the secondary winding open-circuited

The circuit's voltages were probed at the points closest to the real-life frequencies of 100 kHz. Additionally, in order to get the peak-to-peak voltages, the observed voltages (2.2268 and 0.7366 for V_1 and V_2 respectively) were multiplied by 2, as this sort of AC analysis shows the amplitude of the measured voltages, and no DC bias was provided for the circuit. Additionally, the delay could be directly observed, which was 1.5606 degrees or 43.3291 ns. After these datapoints were gathered, they were put into a table of the same form in the Lab 5 document to compare and measure errors more easily (Table 4).

	Measured	Simulated	Error
Frequency (kHz)	100	99.9950	0.0050%
V_1 (V_{pp})	4.44	4.4536	0.3054%
V_2 (V_{pp})	1.44	1.4732	2.2536%
Effective Turns Ratio (V_2/V_1)	3.0833	3.0231	1.9914%
Phase Delta (deg)	1.5120	1.5606	3.1133%

Table 4: Gathered data points and respective errors for the second winding open-circuit circuit

By inspecting the error values in Table 4, some small error values were observed. This could be due to measuring errors caused by the unideal tools and human error whilst measuring the required values. Due to the small magnitude of these errors, the first experiment was deemed a success.

After the first part of the experiment was done, the second configuration was to be created. By utilizing a breadboard, a 10 Ohm resistor was connected between the two terminals of the second winding of the toroidal core (Figure 3.1). After this small step was taken, the previous steps in the first part were taken for the input power configuration and acquisition method settings. However, the channels were switched for the voltages of L_1 and L_2, which meant that CH1 displayed V_2 and CH2 displayed V_1. Using these settings, the voltages of the coupled inductors at 100 kHz and 5 Volts peak-to-peak were measured (Figure 3.2) and the delay between the measured waveforms were cursored (Figure 3.3).

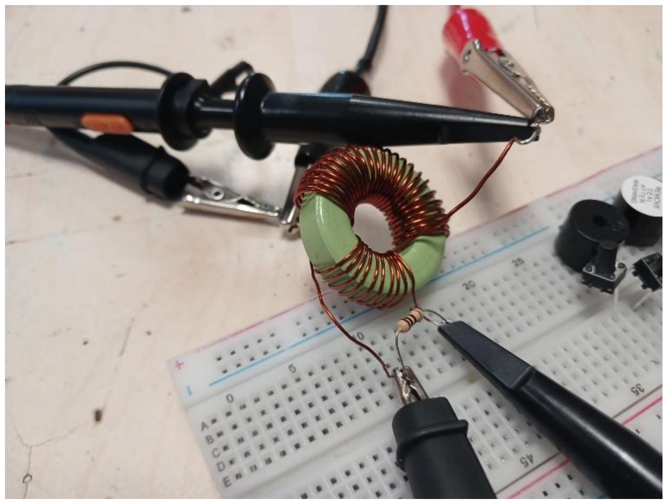

Figure 3.1: Implemented design with loaded second winding

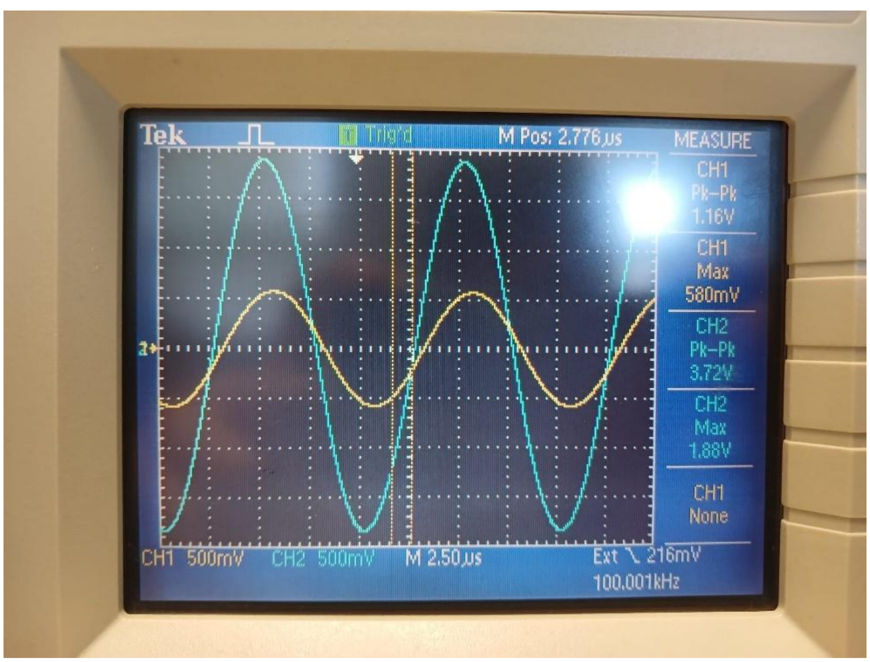

Figure 3.2: Voltages of L_1 $(CH2)$ and L_2 $(CH1)$

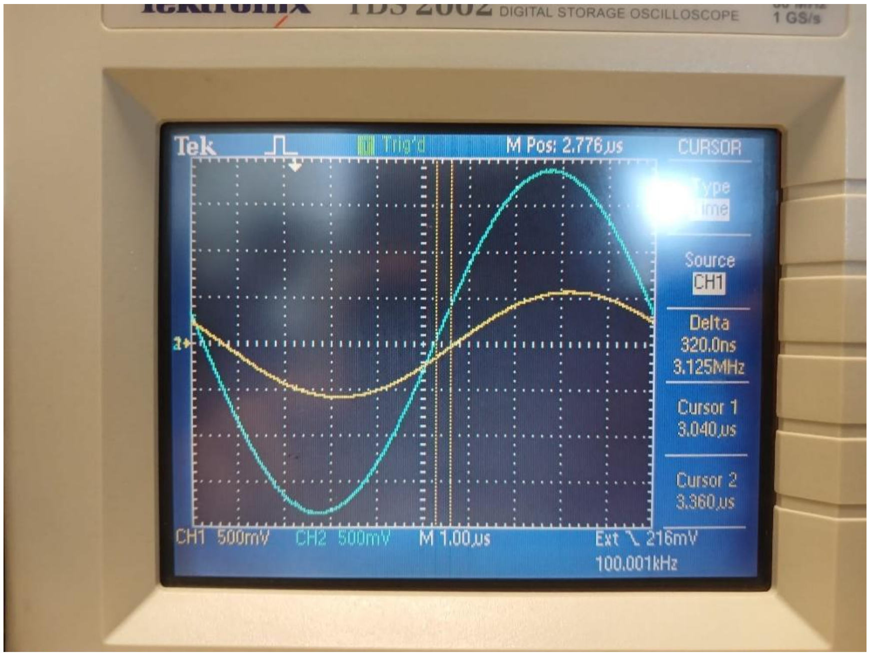

Figure 3.3: Delay between the probed voltages' waveforms

From these images, the peak-to-peak voltages were found to be 3.72 and 1.416 for V_1 and V_2 respectively. Additionally, the delay was found to be approximately 320 ns. When converting this value to degrees, it was multiplied by $\frac{360°}{10\mu s}$, which yielded 11.5200 degrees. These values were taken a note of before further analysis. By utilizing the data in Table 1, the LTSpice model in Figure F was updated. Thei input voltage was set to 5 Volts as an amplitude and a series source resistance of 50 Ohms were added. The simulation was ran and the voltages of the inductors were cursored at the points that included their series resistances (Figure 3.4).

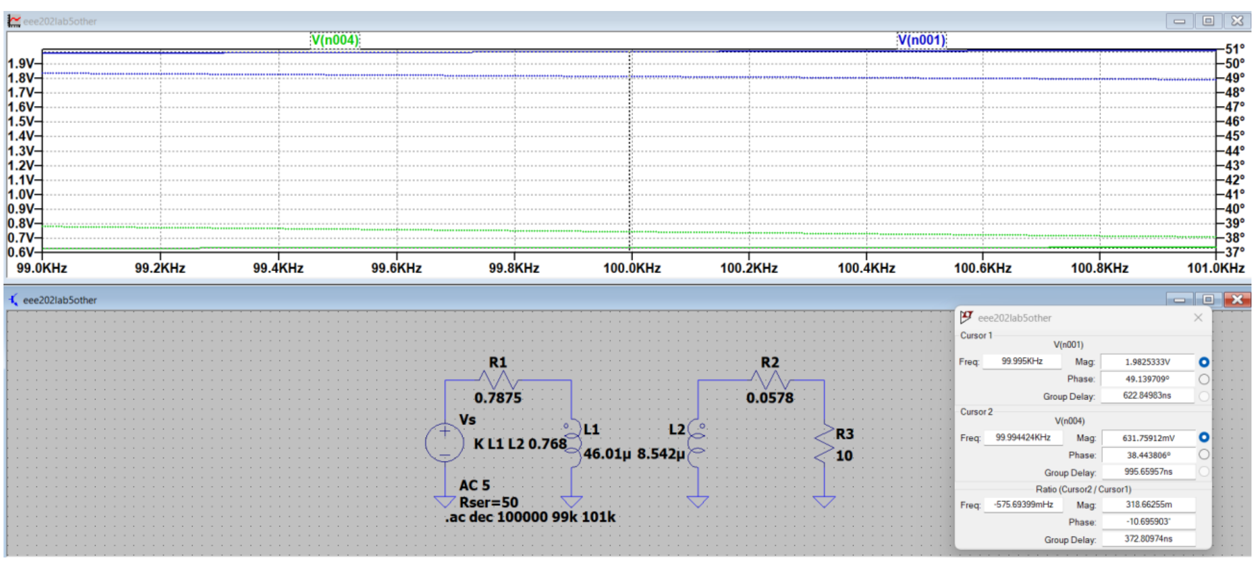

Figure 3.4: Simulation of the real-life circuit with the secondary winding loaded

The circuit's voltages were probed at the points closest to the real-life frequencies of 100 kHz. Additionally, in order to get the peak-to-peak voltages, the observed voltages (1.9825 and 0.6318 for V_1

and V_2 respectively) were multiplied by 2, as this sort of AC analysis shows the amplitude of the measured voltages, and no DC bias was provided for the circuit. Additionally, the delay could be directly observed, which was 10.6959 degrees or 372.8097 ns. After these datapoints were gathered, they were put into a table of the same form in the Lab 5 document to compare and measure errors more easily (Table 5).

	Measured	Simulated	Error
Frequency (kHz)	100	99.9950	0.0050%
V_1 (V_{pp})	3.72	3.965	6.1784%
V_2 (V_{pp})	1.16	1.2636	8.2004%
Effective Turns Ratio (V_2/V_1)	3.2069	3.1378	2.2026%
Phase Delta (deg)	11.5200	10.6959	7.7016%

Table 5: Gathered data points and respective errors for the second winding loaded circuit

By inspecting the error values in Table 5, some error values were observed. This could be due to measuring errors caused by the unideal tools and human error whilst measuring the required values. Compared to the first part, the errors slightly increased, which can be due to extra tolerances brought on by the 10 Ohm load resistor. Due to these facts, the second part was deemed a success.

Conclusion:

The purpose of this lab assignment was to understand mutual inductance and coupled inductors in a better manner. First, the required derivations were made by mathematical manipulation of the given equations in specific configurations -such as coupled and reverse coupled series inductors- to get simple relations that were easy to understand. Then, circuits wherein our derived equations could be tested were implemented using LTSpice in order to test the relations between simulation and theory. For every individual case, a small or no error was observed, confirming our derivations. Through these simulations, mutual inductance and the effect of coupling on series and purely inductive circuits were more deeply understood. Additionally, a specific case wherein the non-source side inductor was loaded was tested. A significant drop in voltage and an extreme increase in phase was observed, which were attributed to the derived equations breaking down when current flows through the inductors. From these results, we have gathered a strong basis for comparing our experimental results to the ideal cases for coupled and reverse coupled series inductive circuits. After the preliminary section, the derived designs were implemented in real-life in order to get real measurements. The components and the measurements yielded some small errors but matched the general preliminary results. Through this laboratory assignment, coupled inductors and mutual inductances were more deeply understood. Additionally, hardware implementation and circuit design skills were gained and improved.

References:

H. Koymen and A. Atalar, *Analog Electronics*, ver. 4.2. Ankara, Turkey: Bilkent University, 2015.